ELEC 231 — Mathematical Methods I for ECE · Fall 2026

Pre-tutorial 2

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Note (Instructions). Work through the problem before your tutorial and upload your solution to onQ as a single PDF by 9:00 AM on the due date (no in-class submission). Show your working, not just the final answer: a genuine, complete attempt earns the 2 marks for this pre-tutorial; a missing or incomplete attempt earns 0.

Problem

Context. Week 2 notes, page 6, "Step response of an RC circuit" (Nilsson & Riedel). A dc current source I_s, a resistor R and a capacitor C are connected as in Figure T2.1; the switch in the top conductor closes at $t = 0$, so that for $t \geq 0$ the three elements are in parallel. The notes give the response

$$v(t) = V_f + (V_0 - V_f)\,\mathrm{e}^{-t/RC}, \qquad t \geq 0,$$

where V_0 is the initial and V_f the final value of the capacitor voltage.

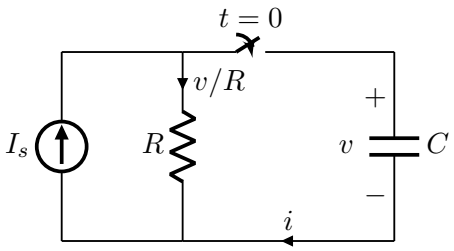


Figure T2.1. The circuit on page 6 of the Week 2 notes. The current i marked on the bottom conductor is the capacitor current i_C; v/R is the resistor current.

Problem 1 (Step response of an RC circuit). Regarding the circuit on page 6 of Week 2 notes (step response of an RC circuit): if $V_0 = 0$, $I_s = 2\,\mathrm{A}$, $R = 1\,\mathrm{k\Omega}$, what is the capacitor current at $t = 0^+$ (switch closed) versus at $t = 0^-$ (switch open). Is there a discontinuity in the capacitor current at $t = 0$, unlike the capacitor voltage?